AGILE DEVELOPMENT AND DEVOPS LAB
ASSIGNMENT-5

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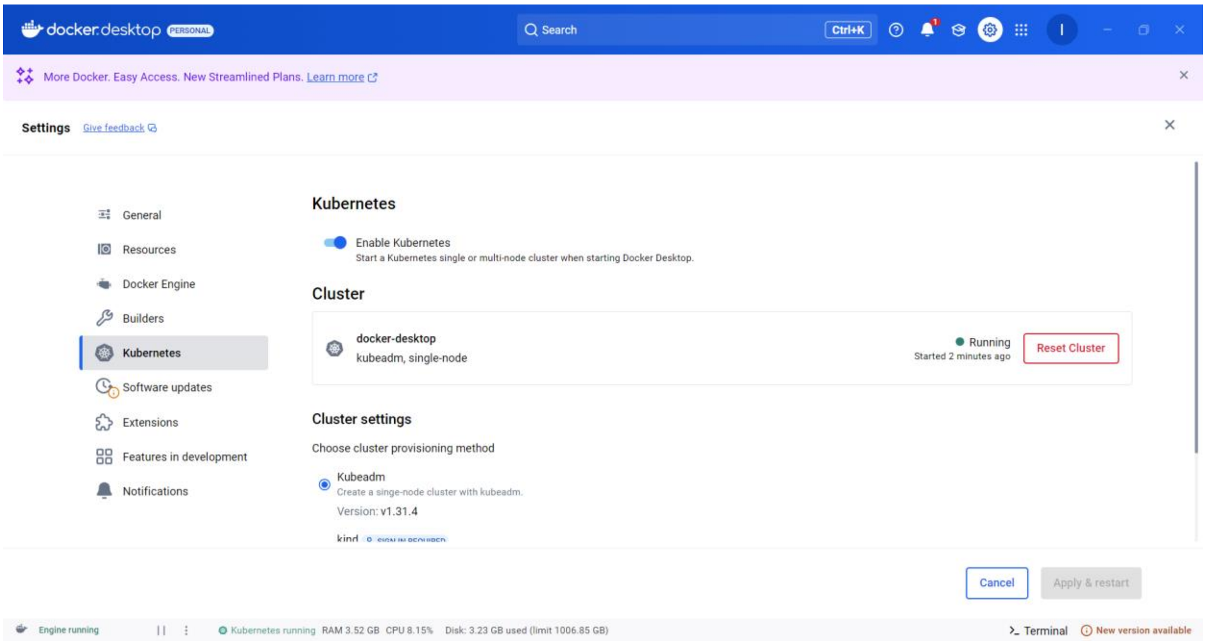
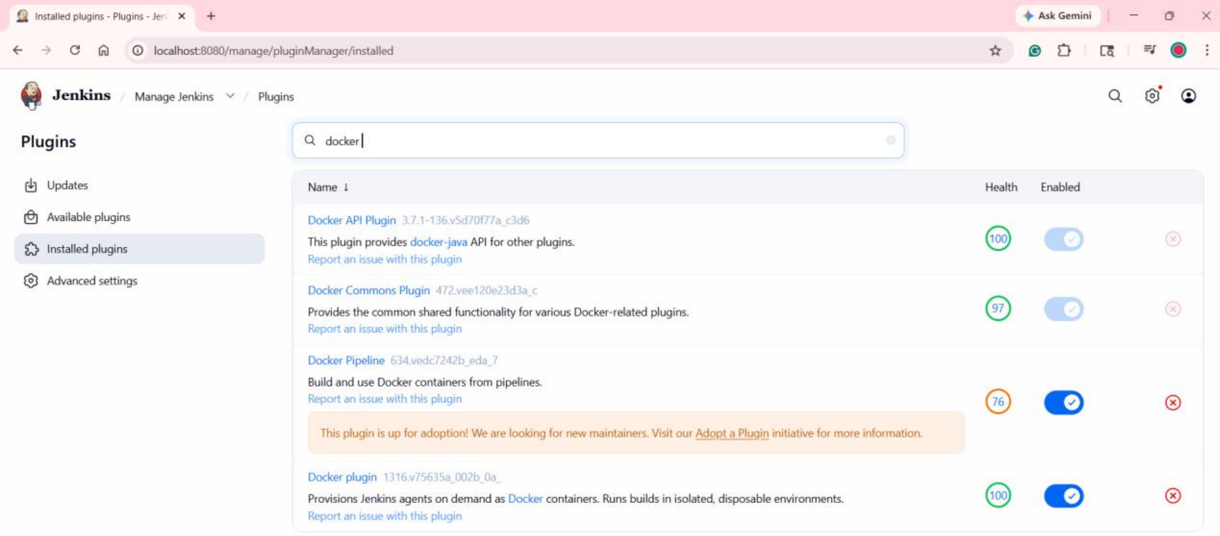
23MIS0072

I. Jenkin – Docker – Kubernetes (CI-CD Demo)

Step1: Install Docker Desktop
Step2: Enable Kubernetes:
Open Docker Desktop Settings -> **Kubernetes** -> Check **Enable Kubernetes**.
This creates a local context named docker-desktop.



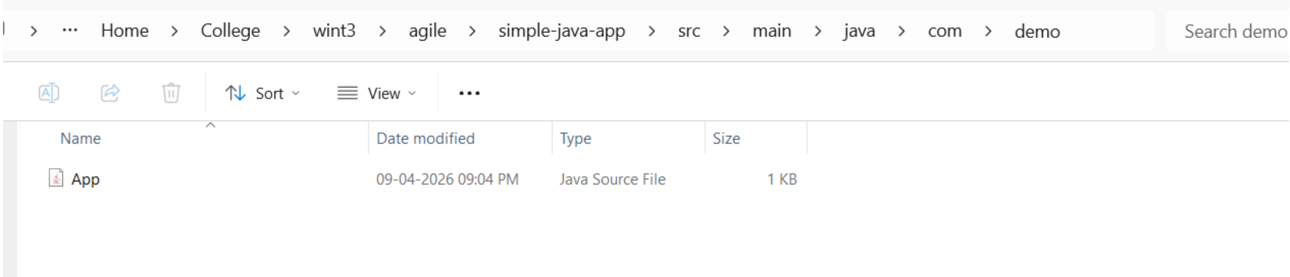
Step 3: In Jenkin, enable two plugins (Docker and Docker Pipeline) and Install them

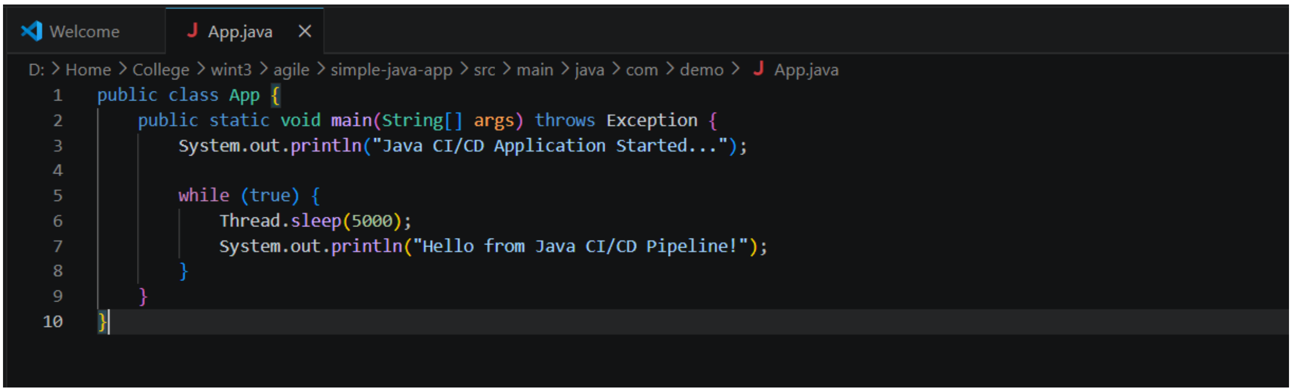


Step 4 : Create Simple Java Application

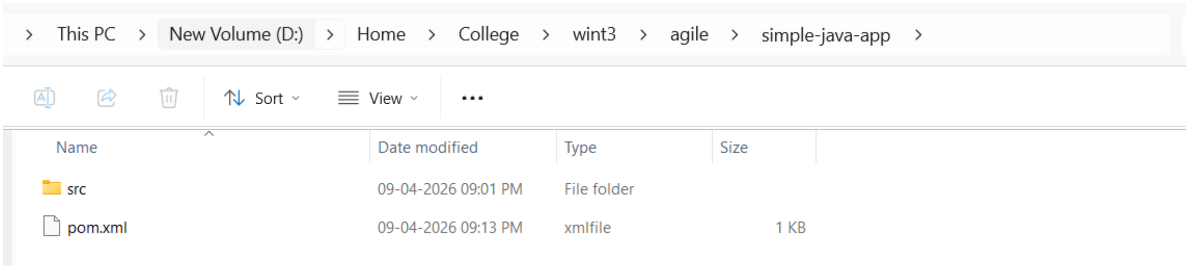
Create a folder in local machine: simple-java-app

Create Java File App.java in the file path “\simple-java-app\src\main\java\com\demo”





Create Maven File pom.xml in ‘simple-java-app’ folder



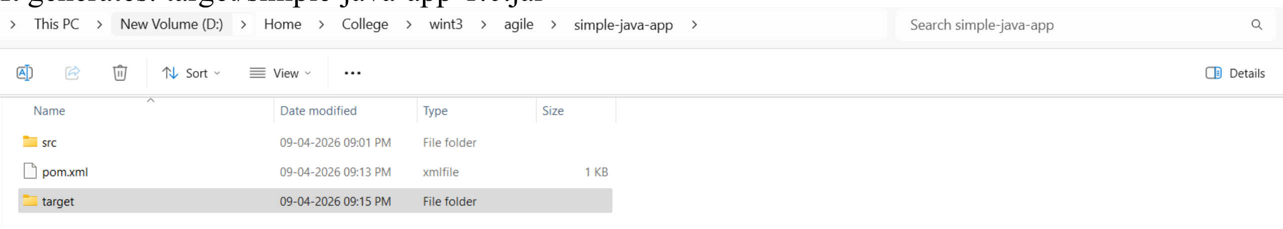
```xml
<project xmlns="http://maven.apache.org/POM/4.0.0"
         xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance"
         xsi:schemaLocation="http://maven.apache.org/POM/4.0.0
         http://maven.apache.org/xsd/maven-4.0.0.xsd">

    <modelVersion>4.0.0</modelVersion>

    <groupId>com.demo</groupId>
    <artifactId>simple-java-app</artifactId>
    <version>1.0</version>

    <build>
        <plugins>
            <plugin>
                <groupId>org.apache.maven.plugins</groupId>
                <artifactId>maven-compiler-plugin</artifactId>
                <version>3.10.1</version>
                <configuration>
                    <source>17</source>
                    <target>17</target>
                </configuration>
            </plugin>
        </plugins>
    </build>
</project>
```

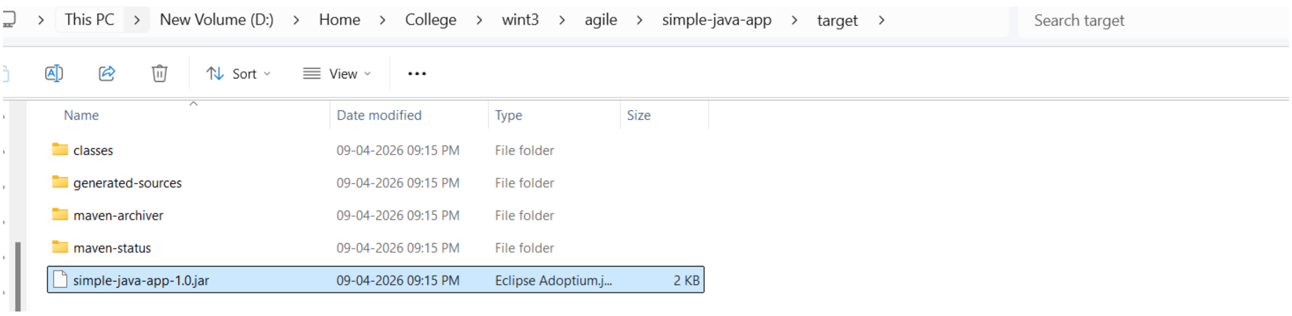
Step 5 - Build Locally (Test)

Inside project folder:

mvn clean package

It generates: target/simple-java-app-1.0.jar






```
PS C:\Users\iffah> cd D:\Home\College\wint3\agile\simple-java-app
PS D:\Home\College\wint3\agile\simple-java-app> mvn clean package
[INFO] Scanning for projects...
[INFO]
[INFO] ---------------------< com.demo:simple-java-app >----------------------
[INFO] Building simple-java-app 1.0
[INFO]   from pom.xml
[INFO] --------------------------------[ jar ]---------------------------------
Downloading from central: https://repo.maven.apache.org/maven2/org/apache/maven/plugins/maven-clean-plugin/3.2.0/maven-clean-plugin-3.2.0.pom
Downloaded from central: https://repo.maven.apache.org/maven2/commons-io/commons-io/2.21.0/commons-io-2.21.0.jar (585 KB at 171 KB
Downloaded from central: https://repo.maven.apache.org/maven2/com/github/luben/zstd-jni/1.5.7-6/zstd-jni-1.5.7-6.jar (7.4 MB at 85
[INFO] Building jar: D:\Home\College\wint3\agile\simple-java-app\target\simple-java-app-1.0.jar
[INFO] ------------------------------------------------------------------------
[INFO] BUILD SUCCESS
[INFO] ------------------------------------------------------------------------
[INFO] Total time:  25.596 s
[INFO] Finished at: 2026-04-09T21:15:55+05:30
[INFO] ------------------------------------------------------------------------
PS D:\Home\College\wint3\agile\simple-java-app>
```

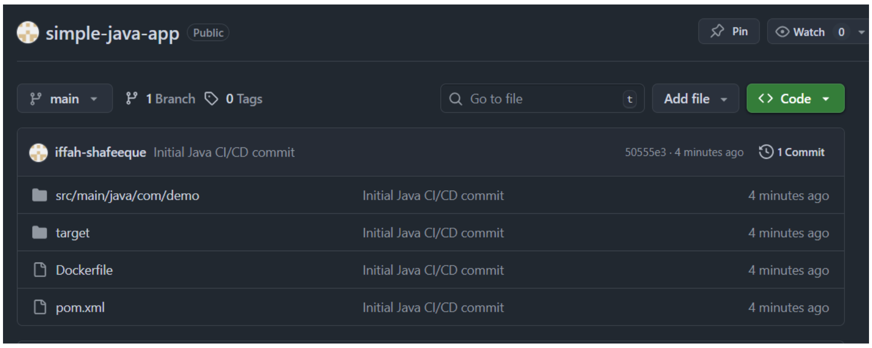
Step 6: Create Dockerfile Dockerfile

```
Welcome   Dockerfile X
D: > Home > College > wint3 > agile > simple-java-app > Dockerfile > ...
1  FROM eclipse-temurin:17-jdk
2  WORKDIR /app
3  COPY target/simple-java-app-1.0.jar app.jar
4  CMD ["java", "-jar", "app.jar"]
```

Step 7 : Push Code to GitHub (Before that create a public repository in the Github)

git init
git add .
git commit -m "Initial Java CI/CD commit
git branch -M main
git remote add origin https://github.com/iffah-shafeeque/simple-java-app
git push -u origin main

```
PS D:\Home\College\wint3\agile\simple-java-app> git init
Initialized empty Git repository in D:/Home/College/wint3/agile/simple-java-app/.git/
PS D:\Home\College\wint3\agile\simple-java-app> git add .
warning: in the working copy of 'target/maven-status/maven-compiler-plugin/compile/default-compile/createdFiles.lst', LF will be replaced by CRLF the next time
Git touches it
warning: in the working copy of 'target/maven-status/maven-compiler-plugin/compile/default-compile/inputFiles.lst', LF will be replaced by CRLF the next time Gi
t touches it
PS D:\Home\College\wint3\agile\simple-java-app> git commit -m "Initial Java CI/CD commit"
[master (root-commit) 50555e3] Initial Java CI/CD commit
 8 files changed, 46 insertions(+)
 create mode 100644 Dockerfile
 create mode 100644 pom.xml
 create mode 100644 src/main/java/com/demo/App.java
 create mode 100644 target/classes/App.class
 create mode 100644 target/maven-archiver/pom.properties
 create mode 100644 target/maven-status/maven-compiler-plugin/compile/default-compile/createdFiles.lst
 create mode 100644 target/maven-status/maven-compiler-plugin/compile/default-compile/inputFiles.lst
 create mode 100644 target/simple-java-app-1.0.jar
PS D:\Home\College\wint3\agile\simple-java-app> git branch -M main
PS D:\Home\College\wint3\agile\simple-java-app> git remote add origin https://github.com/iffah-shafeeque/simple-java-app
PS D:\Home\College\wint3\agile\simple-java-app> git push -u origin main
Enumerating objects: 22, done.
Counting objects: 100% (22/22), done.
Delta compression using up to 8 threads
Compressing objects: 100% (11/11), done.
Writing objects: 100% (22/22), 3.50 KiB | 358.00 KiB/s, done.
Total 22 (delta 0), reused 0 (delta 0), pack-reused 0 (from 0)
To https://github.com/iffah-shafeeque/simple-java-app
 * [new branch]      main -> main
branch 'main' set up to track 'origin/main'.
PS D:\Home\College\wint3\agile\simple-java-app> 
```



Step 8: Create Kubernetes Deployment File *deployment.yaml*

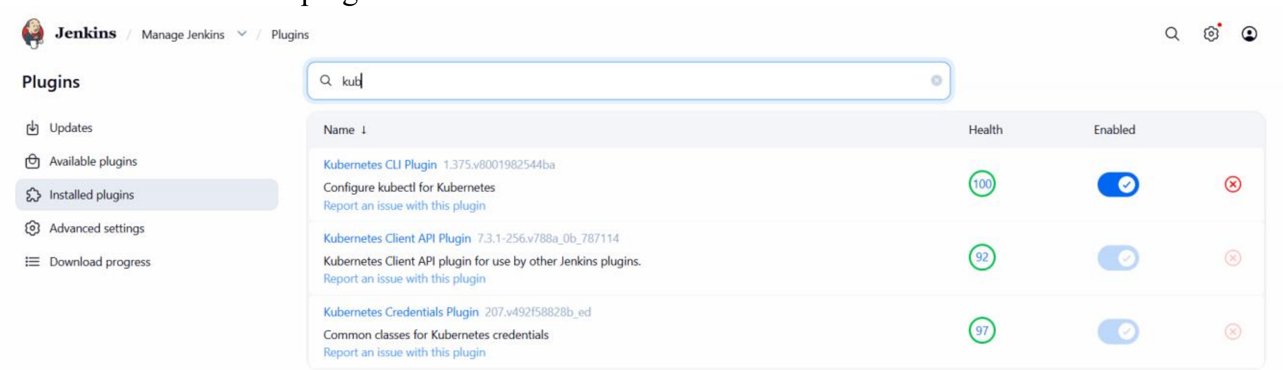
```yaml
apiVersion: apps/v1
kind: Deployment
metadata:
  name: java-app
spec:
  replicas: 2
  selector:
    matchLabels:
      app: java-app
  template:
    metadata:
      labels:
        app: java-app
    spec:
      containers:
        - name: java-app
          image: iffah11/java-app:latest

---
apiVersion: v1
kind: Service
metadata:
  name: java-service
spec:
  type: NodePort
  selector:
    app: java-app
  ports:
    - port: 80
      targetPort: 8080
      nodePort: 30008
```

After creating it, Push it to GitHub:

git add deployment.yaml
git commit -m "Added Kubernetes deployment file"
git push

```
PS D:\Home\College\wint3\agile\simple-java-app> git add deployment.yaml
PS D:\Home\College\wint3\agile\simple-java-app> git commit -m "Added Kubernetes deployment file"
[main 3500f77] Added Kubernetes deployment file
 1 file changed, 31 insertions(+)
 create mode 100644 deployment.yaml
PS D:\Home\College\wint3\agile\simple-java-app> git push
Enumerating objects: 4, done.
Counting objects: 100% (4/4), done.
Delta compression using up to 8 threads
Compressing objects: 100% (3/3), done.
Writing objects: 100% (3/3), 525 bytes | 175.00 KiB/s, done.
Total 3 (delta 1), reused 0 (delta 0), pack-reused 0 (from 0)
remote: Resolving deltas: 100% (1/1), completed with 1 local object.
To https://github.com/iffah-shafeeque/simple-java-app
   50555e3..3500f77  main -> main
PS D:\Home\College\wint3\agile\simple-java-app>
```

Step 9: Configure Jenkins
Run Jenkins
Install Kubernetes CLI plugin



Step 10: Add DockerHub Credentials
In CI/CD:

1. Jenkins builds Docker image
2. Jenkins pushes image to Docker Hub
3. Kubernetes pulls image from Docker Hub

Without credentials → Push will fail

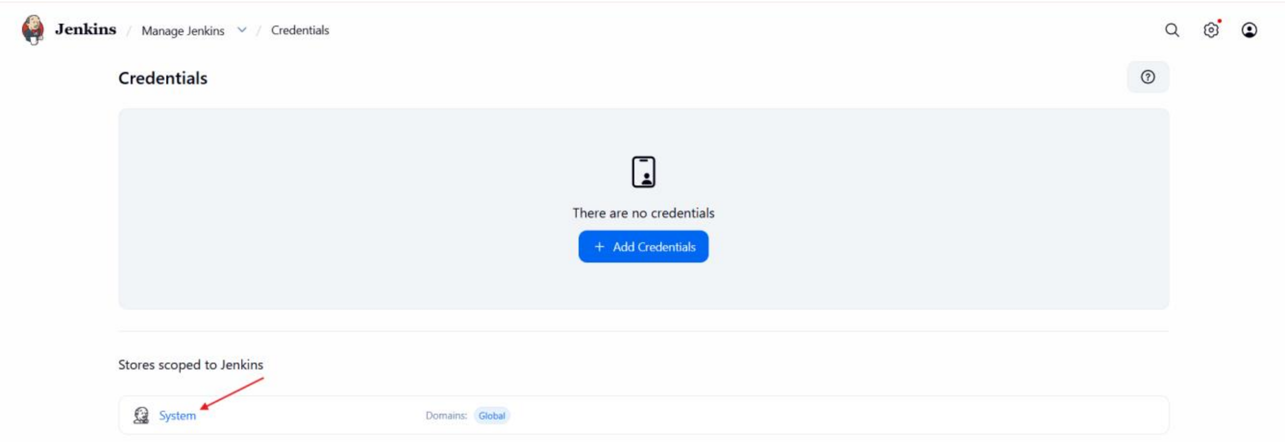
Steps

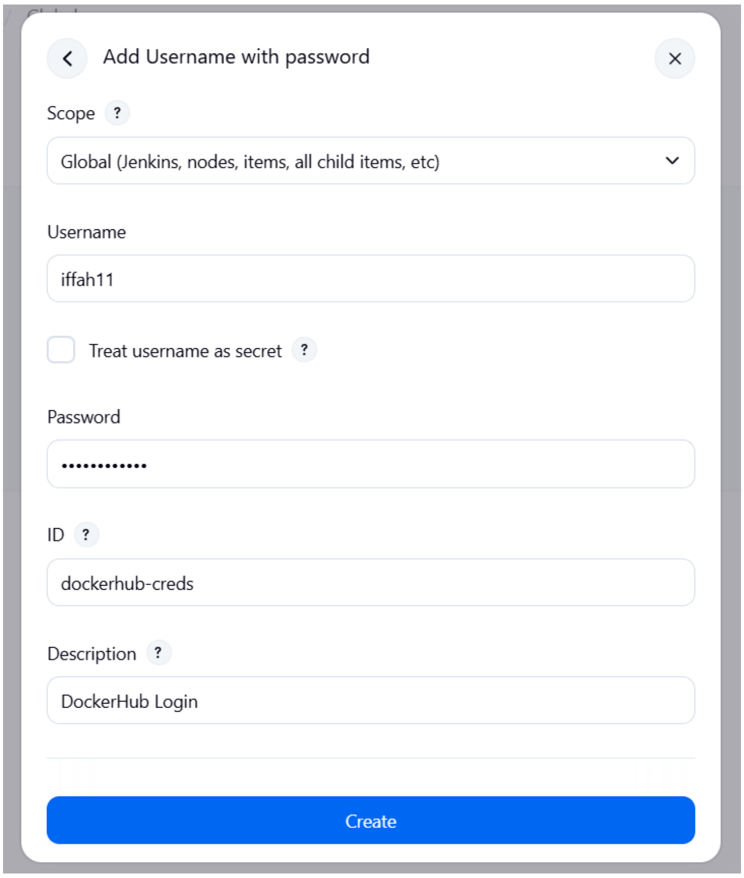
Manage Jenkins -> Credentials ->(System)->Global credentials (unrestricted)->Add Credentials

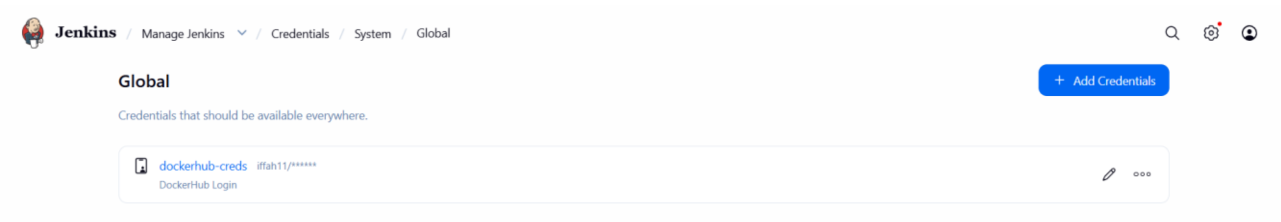
Fill like this:

Field	Value
Kind	Username with password
Scope	Global
Username	**your DockerHub username**
Password	**your DockerHub password**
ID	dockerhub-creds
Description	DockerHub Login

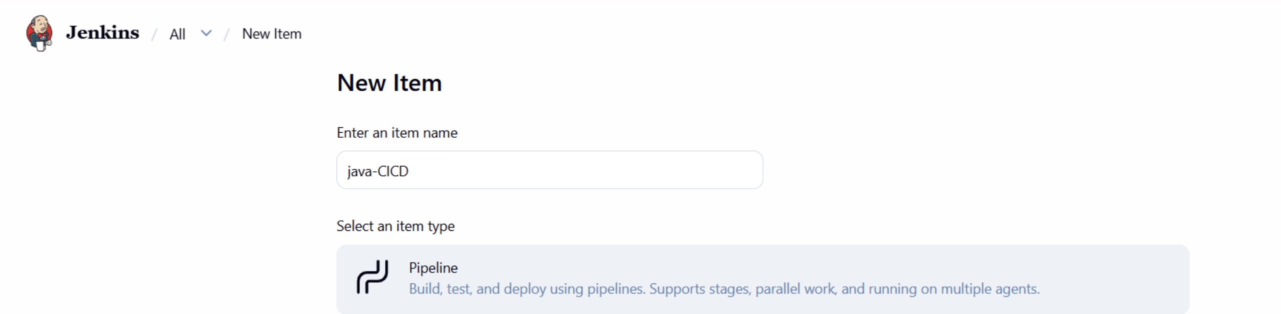
Click Create







Step 11: Create Jenkins Pipeline Job



Pipleline Script:

Script ?

```
pipeline {
    agent any

    environment {
        DOCKER_IMAGE = "iffah11/java-app:latest"
    }

    stages {
        stage('Clone Repository') {
            steps {
                git branch: 'main',
                    url: 'https://github.com/iffah-shafeeque/simple-java-app'
            }
        }
```

```
        stage('Build with Maven') {
            steps {
                bat 'mvn clean package'
            }
        }

        stage('Build Docker Image') {
            steps {
                bat "docker build -t %DOCKER_IMAGE% ."
            }
        }

        stage('Push to DockerHub') {
            steps {
```

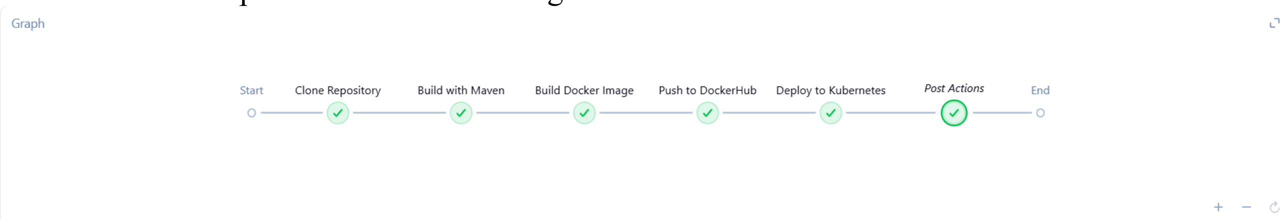
```

                withCredentials([usernamePassword(
                    credentialsId: 'dockerhub-creds',
                    usernameVariable: 'DOCKER_USER',
                    passwordVariable: 'DOCKER_PASS'
                )]) {
                    bat """
                    docker login -u %DOCKER_USER% -p %DOCKER_PASS%
                    docker push %DOCKER_IMAGE%
                    """
                }
            }
        }

        stage('Deploy to Kubernetes') {
```

```
            steps {
                bat 'kubectl apply -f deployment.yaml'
            }
        }
    }

    post {
        success {
            echo 'CI/CD Pipeline executed successfully!'
        }
        failure {
            echo 'Pipeline failed. Please check the Console Output for errors.'
        }
    }
}
```

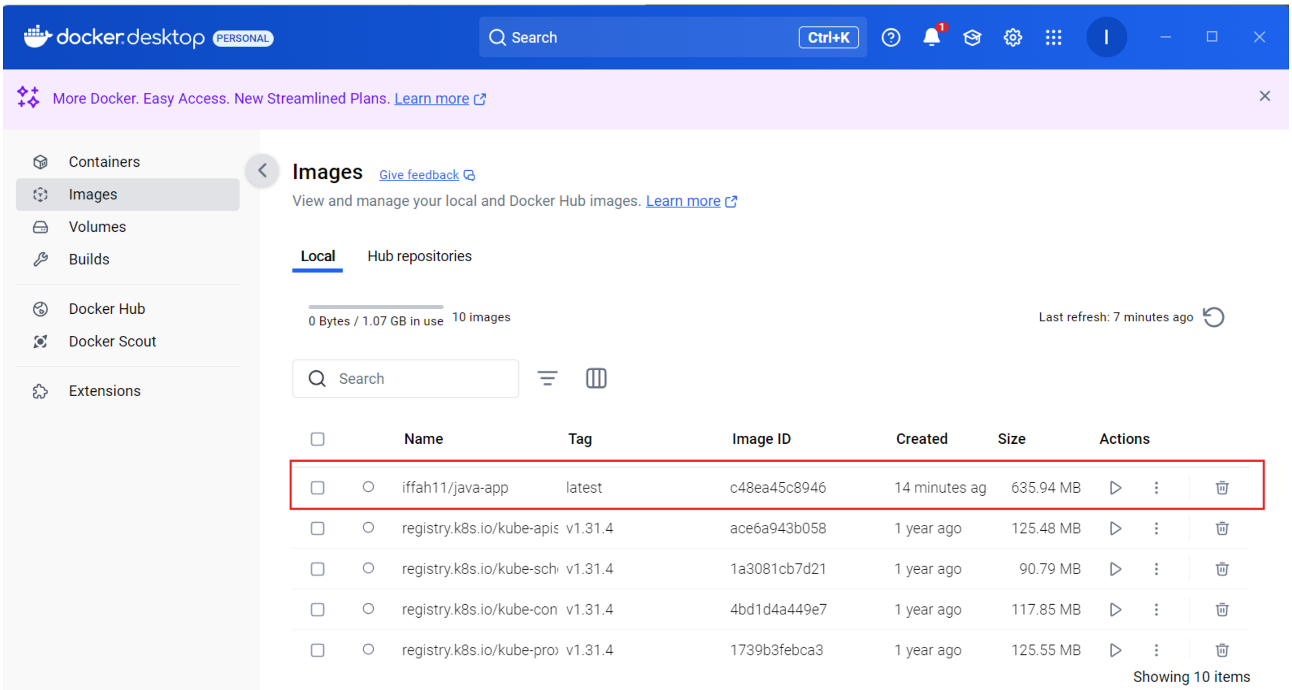
Click → Save → Build Now

```
deployment.apps/java-app created
service/java-service created
[Pipeline] }
[Pipeline] // stage
[Pipeline] stage
[Pipeline] { (Declarative: Post Actions)
[Pipeline] echo
CI/CD Pipeline executed successfully!
[Pipeline] }
[Pipeline] // stage
[Pipeline] }
[Pipeline] // withEnv
[Pipeline] }
[Pipeline] // node
[Pipeline] End of Pipeline
Finished: SUCCESS
```

Above console output is the success message



Step 12: Verify Docker and Kubernetes Deployment
Docker Image



Run this command to verify in command prompt,
kubectl get svc

```
PS C:\Users\iffah> kubectl get svc
NAME           TYPE        CLUSTER-IP     EXTERNAL-IP   PORT(S)        AGE
java-service   NodePort    10.96.78.136   <none>        80:30008/TCP   4m30s
kubernetes     ClusterIP   10.96.0.1      <none>        443/TCP        169m
PS C:\Users\iffah>
```

II. Spring Boot CI/CD Pipeline (GitHub + Jenkins + Docker + Kubernetes)

Step 1: Install Tools: Java 17, Maven, Docker Desktop (enable Kubernetes), Jenkins, and Git.
Step 2: Create Spring Boot Project
pom.xml

D: > Home > College > wint3 > agile > SpringProject > SpringProject > pom.xml

```xml
<project xmlns="http://maven.apache.org/POM/4.0.0"
    xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance"
    xsi:schemaLocation="http://maven.apache.org/POM/4.0.0 https://maven.apache.org/xsd/maven-4.0.0.xsd">
    <modelVersion>4.0.0</modelVersion>
    <groupId>com.iffah</groupId>
    <artifactId>app</artifactId>
    <parent>
        <groupId>org.springframework.boot</groupId>
        <artifactId>spring-boot-starter-parent</artifactId>
        <version>3.2.5</version>
    </parent>
    <dependencies>
        <dependency>
            <groupId>org.springframework.boot</groupId>
            <artifactId>spring-boot-starter-web</artifactId>
        </dependency>
    </dependencies>
</project>
```

App.java

pom.xml | App.java ×

D: > Home > College > wint3 > agile > SpringProject > SpringProject > src > main > java > assignment > App.java

```java
package assignment;

import org.springframework.boot.SpringApplication;
import org.springframework.boot.autoconfigure.SpringBootApplication;

@SpringBootApplication
public class App {
    public static void main(String[] args) {
        SpringApplication.run(App.class, args);
    }
}

```

HelloController.java

pom.xml × | App.java | HelloController.java ×

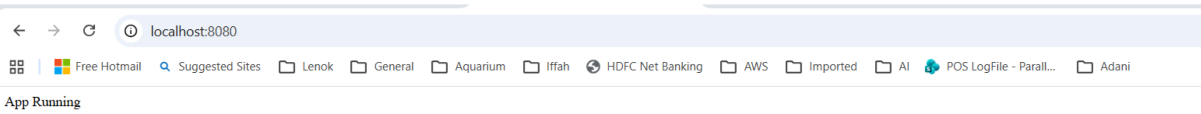
D: > Home > College > wint3 > agile > SpringProject > SpringProject > src > main > java > assignment > HelloController.java

```java
package assignment;

import org.springframework.web.bind.annotation.GetMapping;
import org.springframework.web.bind.annotation.RestController;

@RestController
public class HelloController {
@GetMapping("/")
public String home() { return "App Running"; }
}

```

Step 3: Run Locally

mvn spring-boot:run

```
[INFO] Copying 0 resource from src\main\resources to target\classes
[INFO]
[INFO] --- maven-compiler-plugin:3.11.0:compile (default-compile) @ app ---
[INFO] Nothing to compile - all classes are up to date
[INFO]
[INFO] --- maven-resources-plugin:3.3.1:testResources (default-testResources) @ app ---
[INFO] Copying 0 resource from src\test\resources to target\test-classes
[INFO]
[INFO] --- maven-compiler-plugin:3.11.0:testCompile (default-testCompile) @ app ---
[INFO] Nothing to compile - all classes are up to date
[INFO]
[INFO] <<< spring-boot-maven-plugin:3.2.5:run (default-cli) < test-compile @ app <<<
[INFO]
[INFO]
[INFO] --- spring-boot-maven-plugin:3.2.5:run (default-cli) @ app ---
[INFO] Attaching agents: []

  .   ____          _            __ _ _
 /\\ / ___'_ __ _ _(_)_ __  __ _ \ \ \ \
( ( )\___ | '_ | '_| | '_ \/ _` | \ \ \ \
 \\/  ___)| |_)| | | | | || (_| |  ) ) ) )
  '  |____| .__|_| |_|_| |_\__, | / / / /
 =========|_|==============|___/=/_/_/_/
 :: Spring Boot ::                (v3.2.5)

2026-04-11T22:56:48.717+05:30  INFO 63688 --- [           main] assignment.App                           : Starting App using Java 17.0.12 with PID 63688 (D
:\Home\Iffah\SpringProject\target\classes started by Lenovo in D:\Home\Iffah\SpringProject)
2026-04-11T22:56:48.722+05:30  INFO 63688 --- [           main] assignment.App                           : No active profile set, falling back to 1 default
profile: "default"
2026-04-11T22:56:49.934+05:30  INFO 63688 --- [           main] o.s.b.w.embedded.tomcat.TomcatWebServer  : Tomcat initialized with port 8080 (http)
2026-04-11T22:56:49.956+05:30  INFO 63688 --- [           main] o.apache.catalina.core.StandardService   : Starting service [Tomcat]
2026-04-11T22:56:49.957+05:30  INFO 63688 --- [           main] o.apache.catalina.core.StandardEngine    : Starting Servlet engine: [Apache Tomcat/10.1.20]
2026-04-11T22:56:50.055+05:30  INFO 63688 --- [           main] o.a.c.c.C.[Tomcat].[localhost].[/]       : Initializing Spring embedded WebApplicationContex
t
2026-04-11T22:56:50.057+05:30  INFO 63688 --- [           main] w.s.c.ServletWebServerApplicationContext : Root WebApplicationContext: initialization comple
ted in 1269 ms
2026-04-11T22:56:50.556+05:30  INFO 63688 --- [           main] o.s.b.w.embedded.tomcat.TomcatWebServer  : Tomcat started on port 8080 (http) with context p
ath ''
2026-04-11T22:56:50.578+05:30  INFO 63688 --- [           main] assignment.App                           : Started App in 2.32 seconds (process running for
2.709)
```



Step 4: Docker

Dockerfile

FROM eclipse-temurin:17-jdk

WORKDIR /app

COPY target/app.jar app.jar

ENTRYPOINT ["java","-jar","app.jar"]

docker build -t springboot-app .

```
D:\Home\Iffah\SpringProject>docker build -t springboot-app .
[+] Building 48.1s (8/8) FINISHED                                                                          docker:desktop-linux
 => [internal] load build definition from Dockerfile                                                                       0.0s
 => => transferring dockerfile: 145B                                                                                       0.0s
 => [internal] load metadata for docker.io/library/eclipse-temurin:17-jdk                                                  1.1s
 => [internal] load .dockerignore                                                                                          0.0s
 => => transferring context: 2B                                                                                            0.0s
 => [1/3] FROM docker.io/library/eclipse-temurin:17-jdk@sha256:2ff0b91aaa4cb5dc35b4760c37fe064390a486dcd90b361caa72b55d598601f3  46.1s
 => => resolve docker.io/library/eclipse-temurin:17-jdk@sha256:2ff0b91aaa4cb5dc35b4760c37fe064390a486dcd90b361caa72b55d598601f3  0.0s
 => => sha256:a1e6e1a167f1d8e27a6ca55c32fa609912f9ff5b0fc88c77413259b59e72ac45 22.97MB / 22.97MB                           17.8s
 => => sha256:f00bd8c488bba663bd5f60752b0a1ac92bc39a10729a5a8bf7227b10f6096904 2.28kB / 2.28kB                              0.8s
 => => sha256:a7428f4f5c1dad835231adfcff057cad6efc5b2dc256a36871e2e72cb75561e4 145.63MB / 145.63MB                         44.1s
 => => sha256:a149c623348dfa371df99c86694ffdf86087479170658504 5a923ea4e6224b0d 158B / 158B                                 1.1s
 => => sha256:689b91d88a0f4086057ec826027b128902ecf2b516be510371c115bc55da19a6 29.73MB / 29.73MB                           30.6s
 => => extracting sha256:689b91d88a0f4086057ec826027b128902ecf2b516be510371c115bc55da19a6                                  1.1s
 => => extracting sha256:a1e6e1a167f1d8e27a6ca55c32fa609912f9ff5b0fc88c77413259b59e72ac45                                  0.7s
 => => extracting sha256:a7428f4f5c1dad835231adfcff057cad6efc5b2dc256a36871e2e72cb75561e4                                  1.8s
 => => extracting sha256:a149c623348dfa371df99c86694ffdf860874791706585045a923ea4e6224b0d                                  0.0s
 => => extracting sha256:f00bd8c488bba663bd5f60752b0a1ac92bc39a10729a5a8bf7227b10f6096904                                  0.0s
 => [internal] load build context                                                                                          0.0s
 => => transferring context: 2.66kB                                                                                        0.0s
 => [2/3] WORKDIR /app                                                                                                     0.4s
 => [3/3] COPY target/app.jar app.jar                                                                                      0.0s
 => exporting to image                                                                                                     0.2s
 => => exporting layers                                                                                                    0.1s
 => => exporting manifest sha256:2280f9753ec2e39f57fd8b09cad05a950d973b3cb621a6d07b2873383f8e2cd1                          0.0s
 => => exporting config sha256:10f4c5c30c2ea73f92ae602038df712c003fc82818e7439b73dbc57818b22713                            0.0s
 => => exporting attestation manifest sha256:3afe873ccbb428b24aca0daf73c0b69b78d7cd8156ef9ae86536da8dff276ef6              0.0s
 => => exporting manifest list sha256:014337f455d5b6fbc8324b5c5d5bec7fbd52e7c62fdfd00c84af037e3ef00ab4                     0.0s
 => => naming to docker.io/library/springboot-app:latest                                                                   0.0s
 => => unpacking to docker.io/library/springboot-app:latest                                                                0.0s

What's next:
    View a summary of image vulnerabilities and recommendations → docker scout quickview
```

docker run -p 8080:8080 springboot-app

```
D:\Home\Iffah\SpringProject>docker run -p 8080:8080 springboot-app
no main manifest attribute, in app.jar
```

Step 5: Kubernetes

pom.xml | App.java | HelloController.java | deployment.yaml

D: > Home > College > wint3 > agile > SpringProject > SpringProject > deployment.yaml

```yaml
apiVersion: apps/v1
kind: Deployment
metadata:
name: springboot-app
spec:
replicas: 1
template:
spec:
containers:
- name: app
image: springboot-app
imagePullPolicy: Never
```

Step 6: GitHub

```
PS D:\Home\College\wint3\agile\SpringProject\SpringProject> git init
Reinitialized existing Git repository in D:/Home/College/wint3/agile/SpringProject/SpringProject/.git/
PS D:\Home\College\wint3\agile\SpringProject\SpringProject> git add .
warning: in the working copy of 'springboot_cicd_guide (3) (2).pdf', LF will be replaced by CRLF the next time Git touches it
warning: in the working copy of 'target/maven-status/maven-compiler-plugin/compile/default-compile/inputFiles.lst', LF will be replaced by CRLF the next time Git touches i
t
PS D:\Home\College\wint3\agile\SpringProject\SpringProject> git commit -m "init"
On branch master
nothing to commit, working tree clean
```

```
PS D:\Home\College\wint3\agile\SpringProject\SpringProject> git remote add origin https://github.com/iffah-shafeeque/spring
PS D:\Home\College\wint3\agile\SpringProject\SpringProject> git push origin master
Enumerating objects: 43, done.
Counting objects: 100% (43/43), done.
Delta compression using up to 8 threads
Compressing objects: 100% (30/30), done.
Writing objects: 100% (43/43), 9.93 KiB | 1.10 MiB/s, done.
Total 43 (delta 1), reused 0 (delta 0), pack-reused 0 (from 0)
remote: Resolving deltas: 100% (1/1), done.
To https://github.com/iffah-shafeeque/spring
 * [new branch]      master -> master
PS D:\Home\College\wint3\agile\SpringProject\SpringProject>
```

Step 7: Jenkins Pipeline

Jenkins / spring / Configure

Configure

General
Triggers
Pipeline
Advanced

Script

```groovy
pipeline {
    agent any

    stages {
        stage('Clone') {
            steps {
                git 'https://github.com/iffah-shafeeque/spring'
            }
        }

        stage('Build') {
            steps {
                bat 'mvn clean package'
            }
        }

        stage('Docker') {
            steps {
                bat 'docker build -t springboot-app .'
            }
        }

        stage('Deploy') {
            steps {
                bat 'kubectl apply -f deployment.yaml'
            }
        }
    }
}
```

Save Apply

Build 18s

Docker Build 22s

Deploy 2s